Sporozoites are minute thread-like curved organisms with tapering ends. Measuring 9–12μ in length with a central elongated nucleus while, the cytoplasm reveals no pigment as seen with a light microscope. In human, schizogony occurs in two locations. One in the red blood cells (erythrocytic schizogony) and other in the liver cells (pre – or exoerythrocytic schizogony).

A. Pre–erythrocytic or Exoerythrocytic schigony

- Sporozoites do not directly enter the RBC's to initiate erythrocyticschizogony, but undergo developmental phase in other human tissues.
- This cycle lasts for about 8 days in *Plasmodium vivax*, 6 days in *P. falciparum* and 9 days in *P. ovale*.
- This pre–erythrocytic schizogony occurs within parenchymal cells of the liver.
- The Sporozoites, which are elongated spindle – shaped bodies, become rounded inside the liver cells.
- They enlarge in size and undergo repeated nuclear division to form several daughter nuclei, each of which is surrounded by cytoplasm.
- This stage of the parasite is called the pre–erythrocytic or exoerthrocytic schizont or merozoites.
- The heptocyte is distended by the enlarging schizont and the liver cell nucleus is pushed to the periphery.
- Mature liver stage schizonts are spherical multinucleate and contain 2000–50,000 uninucleate merozoites.
- These normally rupture in 6–15 days and release thousands of merozoites into the blood stream.
- They do not return from red blood cells to liver cells.

Plasmodium vivax and *P. ovale* – parasites in liver tissue are called hypnozoites.

B. Erythrocyticstage

- The merozoites released by pre–erythrocytic schizonts invade the red blood cells (Parasitaemia).
- Merozoites are pear – shaped bodies, about 1.5 μm in length.
- In the erythrocyte, the merozoite loses its internal organelles and appears as rounded body having a vacuole in the center with the cytoplasm pushed to the periphery and the nucleus at one pole. These forms are called ring forms or young trophozoites.
- The parasite feeds on the hemoglobin of the erythrocyte. They incompletely metabolize hemoglobin therefore, hematin – globin pigment or haemozoin pigment is left behind.
- The malaria pigment released when the parasitized cells rupture is taken up by recticuloendothelial cells.
- The ring form develops and becomes irregular in shape and shows amoeboid motility. This is called the amoeboid form.
- When the amoeboid form reaches a certain stage of development, its nucleus starts dividing by mitosis followed by a division of cytoplasm to become mature schizonts or merozoites.

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